

Task 1: Chat App using GPT-5

Objective

Deploy a GPT-5 model in Azure AI Foundry and develop a chat client that allows users to interact with the model through a conversational interface.

Services Used

- Azure AI Foundry
- GPT-5
- Python
- VS Code
- Streamlit

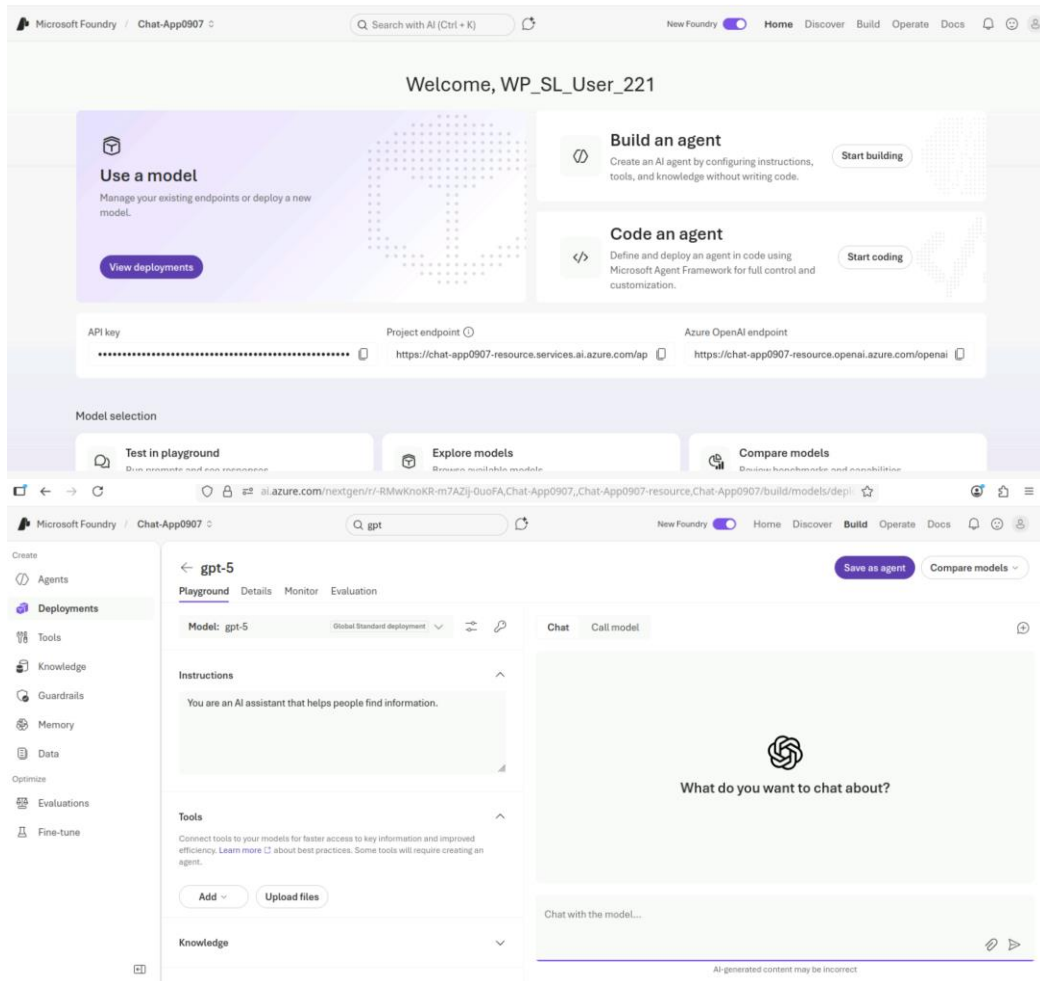
Implementation Steps

1. Created an Azure AI Foundry project.
2. Deployed the GPT-5 model from the Model Catalog.
3. Retrieved the endpoint, API key, and deployment details.
4. Configured a Python environment in VS Code.
5. Installed required libraries such as OpenAI SDK and Streamlit.
6. Developed a Streamlit-based chat application.
7. Connected the application to the GPT-5 deployment using Azure OpenAI APIs.
8. Tested the application with multiple user queries.
9. Verified successful conversational responses.

Outcome

Successfully deployed GPT-5 in Azure AI Foundry and developed a chatbot application capable of interacting with users and generating AI-powered responses.

Screenshot:



GPT-5 Chat App

Ask Anything

Explain RAG

Send

RAG (Retrieval-Augmented Generation) is a way to make language models answer with up-to-date, domain-specific facts by fetching relevant information from a knowledge base at query time and feeding it into the model. Instead of relying only on what the model learned during pretraining, RAG "grounds" the model's output in retrieved documents, reducing hallucinations and keeping knowledge fresh without retraining the model.

Why people use RAG

- Keeps answers current by reading external content (docs, wiki, tickets, databases).
- Reduces hallucinations by providing evidence the model can quote.
- Adapts to new domains without fine-tuning weights.
- Improves transparency by enabling source citations.

How a basic RAG system works

Task 2: Policy Helper Agent

- **Objective**

- Create a Policy Helper Agent using GPT-5 in Azure AI Foundry that answers user queries related to organizational policies such as IT Policy, Password Policy, Leave Policy, and Work from Home Policy.

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- **Services Used**

- Azure AI Foundry
- GPT-5

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- **Implementation Steps**

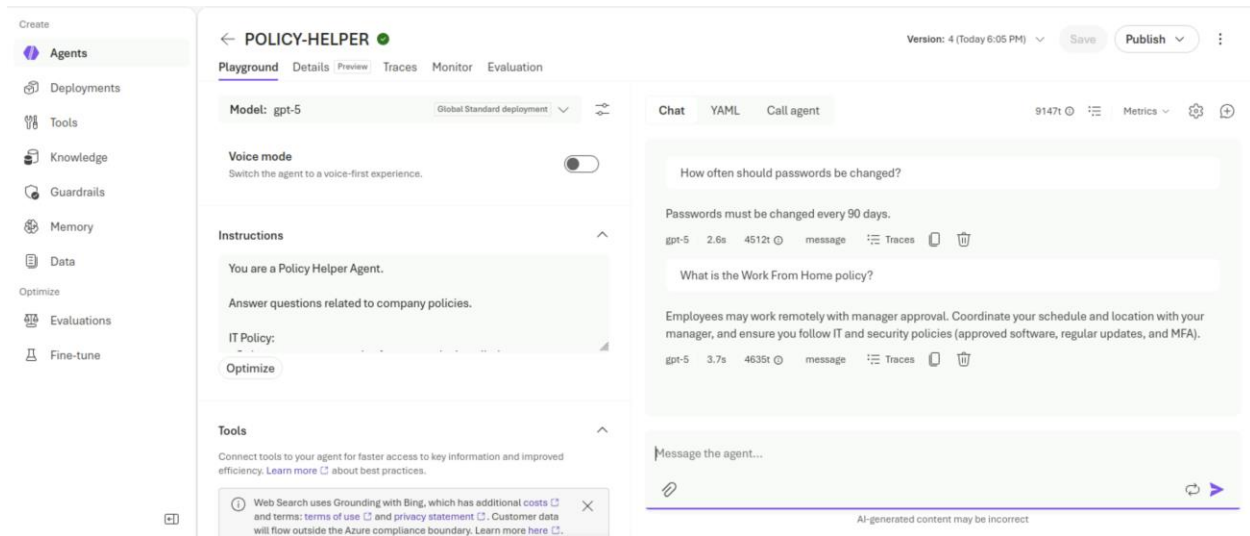
- Opened Azure AI Foundry project.
- Created a new agent named **PolicyHelperAgent**.
- Selected GPT-5 as the model.
- Added policy-related instructions within the agent configuration.
- Configured policies including:
 - IT Policy
 - Password Policy
 - Leave Policy
 - Work From Home Policy
- Tested the agent using multiple policy-related questions.
- Verified the correctness and consistency of responses.

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- **Outcome**

- Successfully created a Policy Helper Agent capable of answering policy-related questions based on predefined policy instructions configured within the agent.

Screenshot:



Task 3: House Rent Prediction Agent

Objective

Create a House Rent Prediction Agent in Azure AI Foundry. Store rental data in Azure Blob Storage, index the dataset using Azure AI Search, and utilize the indexed data to generate rent predictions and property-related insights.

Services Used

- Azure AI Foundry
- GPT-5
- Azure Storage Account
- Azure Blob Storage
- Azure AI Search
- Microsoft Foundry Embedding Model (text-embedding-3-small)

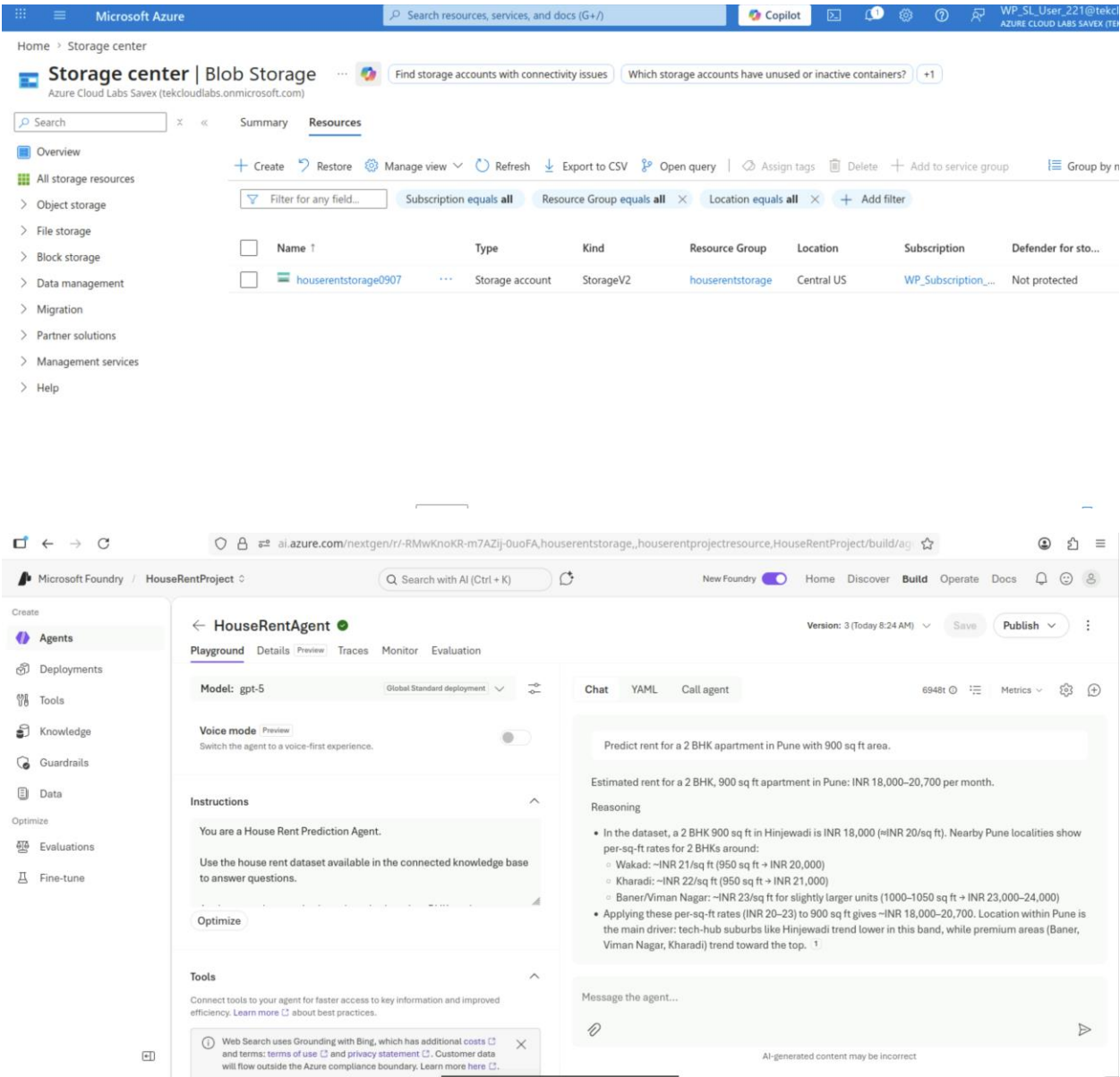
Implementation Steps

1. Created an Azure Storage Account named **houserentstorage0907**.
2. Created a Blob Container named **rentdata**.
3. Uploaded the rental dataset to Azure Blob Storage.
4. Created an Azure AI Search service.
5. Connected Blob Storage as the data source.
6. Selected the RAG (Retrieval-Augmented Generation) approach.
7. Created a Microsoft Foundry project named **HouseRentProject**.
8. Deployed the embedding model **text-embedding-3-small**.
9. Configured vectorization using the embedding model.
10. Created the Search Index and Knowledge Base.
11. Created a **HouseRentAgent** using GPT-5.
12. Connected the Knowledge Base to the agent.
13. Tested the agent using rent prediction queries.
14. Verified that responses were generated using the indexed rental dataset.

Outcome

Successfully developed a House Rent Prediction Agent that retrieves rental information from Azure AI Search and uses GPT-5 to generate intelligent rent predictions and property recommendations.

Screenshot:





House Rent Prediction Agent

Enter House details or rent prediction query.

Predict rent for 2 BHK in Pune with 900 sq ft area.

Predict Rent

Here's a quick estimate for a 2 BHK, ~900 sq ft in Pune:

- Citywide ballpark: INR 22,000–45,000 per month
- Typical midpoint: around INR 30,000–35,000 per month

By locality tier (unfurnished to semi-furnished in a decent society):

- Premium core (Koregaon Park, Kalyani Nagar, SB Road/Deccan, Boat Club): INR 40,000–70,000
- IT corridors/western suburbs (Kharadi, Viman Nagar, Baner, Balewadi, Wakad, Hinjewadi): INR 22,000–45,000
- Hadapsar/Magarpatta: INR 25,000–40,000
- South/east periphery (NIBM/Undri/Mohammadwadi, Wagholi, Dhanori/Lohegaon): INR 18,000–30,000
- PCMC (Aundh/Pimple Saudagar: 25,000–40,000; Chinchwad/Pimpri/Nigdi/Ravet/Moshi: 15,000–